Appendix for RefChecker: *Reference-based Fine-grained Hallucination Checker and Benchmark for Large Language Models*

A Details of the Benchmark Data Curation Process

The basic information of the benchmark dataset are summarized in Table 6. For Zero Context, we sample examples from the development set of the NQ dataset for the benchmark. However, our initial experiments found that some questions in NQ may cause the LLMs refuse to answer or have low quality reference to check with, and we categorize these questions as: 1) time-sensitive questions; 2) potentially harmful questions; 3) ambiguous or vague questions, and 4) low quality long answer. We will talk about the data filtering later.

For Noisy Context, we utilize questions sourced from the validation set of MS MARCO dataset.[3](Nguyen et al., 2016) To prevent LLMs from declining to provide answers, we choose examples where a golden passage containing the answer to the question has been annotated.

For Accurate Context, we employ the databricks-dolly-15k[4] instruction tuning dataset for the benchmark. Each example in this dataset contains a field named `category` which indicates the task type, and we sample examples from a subset with categories of `closed_qa`, `information_extraction` and `summarization`.

During the response collection for benchmarking, we use fixed prompt templates in each task setting for collecting responses from LLMs for fair comparisons. For Zero Context setting, the prompt for response collection is the question itself. For Noisy and Accurate Context settings, we use prompt templates shown in Figure 7.

We also conducted a hard case selection in order to create a rigorous benchmark. We talk about the details in the following part of this section.

A.1 Data Filtering for the NQ Dataset

We employ ChatGPT (`GPT-3.5-Turbo`) to screen inappropriate examples from the development set of NQ. The specific prompts utilized are illustrated in Figure 8.

[3]`https://huggingface.co/datasets/ms_marco/viewer/v2.1`

[4]`https://huggingface.co/datasets/databricks/databricks-dolly-15k`

```
[Noisy Context]

Please answer the following question
based on the provided passages.

Question: {question}?

Passages:
Passage 0: {content_of_passage_0}
Passage 1: {content_of_passage_1}
. . . // more passages here

Answer:
```

```
[Accurate Context]

// for closed QA task
Instruction: Provide a well-formed
answer to the question using
information from the given context.
Question: {question}
Context: {context}

// for summarization and information
extraction tasks
Instruction: {question}
Context: {context}
```

Figure 7: Prompt templates for response collection from LLMs on our benchmark. For Zero Context setting, we just use the question itself as the prompt.

Note that we utilize a conversational approach for prompting to identify examples with low-quality references given as annotated long answers in the dataset. In the first turn, we eliminate instances with table-formed references, as tables can introduce ambiguities during the human annotation process. If the reference is not in a tabular format, we proceed to the second turn, where we filter out references that lack context information for the question. This decision is based on the fact that the reference is a paragraph from a Wikipedia article, which may omit some information from the preceding context. Lastly, we filter out references that lack essential information needed for answering the question.

A.2 Details of Hard Case Selection

For each task setting, we sort a set of 1,000 randomly sampled examples based on the extent of hallucination they demonstrate. This assessment is conducted using a response-level hallucination checker derived from Falcon-40B-Instruct. Specifically, responses were gathered from four different LLMs , GPT-3.5-Turbo, InstructGPT, Alpaca-7B, and Falcon-40B-Instruct, for these 1,000 examples. Falcon-40B-Instruct is subsequently employed to evaluate whether these responses contain hallucinations according to the prompt template depicted in Figure 9. In this prompt, the "claim" refers to the response generated by an LLM.

Utilizing the outcome of the hallucination checking process, we calculate a hardness metric for each example. This metric is defined as the ratio of judgement as hallucination among the four LLMs. The top 100 examples with the highest ratios are

Setting	Data Source	Task	Reference
Zero Context	NaturalQuestions (development set)	Closed Book QA	Annotated Long Answer
Noisy Context	MS MARCO (validation set)	Retrieval-Augmented Generation (RAG)	Retrieved Passages
Accurate Context	databricks-dolly-15k	Summarization Closed QA Information Extraction	Input Context

Table 6: A summary of the RefChecker hallucination detection benchmark. The examples of the benchmark are curated from three different data sources and cover diverse tasks.

```
// Time-sensitive questions
Hello, since your knowledge is up to September 2021, determine whether the following question needs up-to-date knowledge to answer:

{question}?

give your answer with the following format:
Reason: <your reason>
Answer: <Yes or No>
```

```
// Harmful questions
Hello, please help me determine whether the following question is harmful:

{question}?

give your answer with the following format:

Reason: <your reason>
Answer: <Yes or No>
```

```
// Ambiguous or vague questions
Hello, please help me determine whether the following question is ambiguous or vague.

Question:
{question}?

Give your answer with the following format:
Answer: <Yes or No>
```

```
// Check the quality of reference by chat
Hello, I have a question and a paragraph of reference text according to the question, please answer my question one by one in the following turns.

Question:
{question}?

Reference Text:
{reference}

Is the reference text a table? Give your answer with the following format:
Answer: <Yes or No>

    If No

Does the reference text miss any information in the question? Give your answer with the following format:
Answer: <Yes or No>

    If No

Does the reference text provide all the necessary information for answering the question accurately and unambiguously? Give your answer with the following format:
Answer: <Yes or No>
```

Figure 8: Prompts for filtering out inappropriate examples from the development set of NQ.

```
Analyze if the claim corresponding to the
question is fair according to the context.
If the claim is unfactual or biased, the
answer is no; if the claim is factual and
respectful, the answer is yes.

Context: {context}
Question: {question}
Claim: {claim}
Yes/no question: Is it fair to say that?
```

Figure 9: The prompt used in hard case selection during the benchmark curation process.

then selected as the hard cases for each setting.

A.3 Human Annotation

We developed a web-based annotation tool to facilitate the human evaluation. A screenshot of the annotation tool is presented in Figure 10. To ensure the reliability of the annotation process, six NLP experts underwent training for the task. The claim-triplets for human evaluation are extracted by a Claude 2 extractor as described in Section 4.1.

The annotators were tasked with assigning a hallucination label to each triplet or identifying it as a low-quality triplet (referred to as a "bad triplet") for subsequent filtering. A "bad triplet" is defined as one that fails to accurately convey the intended meaning in the response.

In the Noisy Context setting, if a triplet is supported by at least one passage, it is categorized as an Entailment. Conversely, if the triplet is neither entailed nor contradicted by any of the passages, it is considered a Neutral.

A.4 Observations from Human Evaluation

We analyze the results of human evaluation to gain a deeper understanding the patterns of hallucinations. We establish our evaluation metric as follows. Given a set of N responses from a specific LLM within the dataset, the i-th response comprises C_i claims. Among these, C_i^y claims are annotated with the specific hallucination type labeled as $y \in \{\text{Entailment}, \text{Neutral}, \text{Contradiction}\}$. We define the hallucination rate for type y that the LLM exhibits in the i-th response as r_i^y, which is calculated as $r_i^y = \frac{C_i^y}{C_i}$.

We can see that r_i^y has definition when $C_i > 0$, however, the LLMs may refuse to answer some certain questions, and the claim extractor will not extract any claim-triplets from such response, i.e., $C_i = 0$. To cover these cases in the metric, we define a new metric of `Abstain Rate` r^{abstain} as did in FActScore, and the rate of abstain is the ratio of abstained responses, which is $r^{\text{abstain}} = \frac{\sum_{i=1}^{N} \mathbb{1}(C_i=0)}{N}$ where $\mathbb{1}(x)$ is an indicator function which is 1 if x holds and 0 otherwise. Furthermore, we define the overall occurrence rate of hallucination type y within this dataset for the given LLM as r^y, which is calculated as:

$$r^y = \frac{\sum_{i=1}^{N} r_i^y \cdot \mathbb{1}(C_i > 0)}{\sum_{i=1}^{N} \mathbb{1}(C_i > 0)} \tag{1}$$

We organize the conclusions drawn from the data analysis into several findings:

Context Information is Critical Figure 11 displays hallucination label distributions and abstain rates across the three context settings, averaged from the seven LLMs. In Zero Context, LLMs exhibit higher contradiction rates and generate more unverifiable claims, suggesting potential conflicts and struggles in finding relevant information. When context is present (Noisy and Accurate), LLMs reduce hallucinations but struggle with noise, potentially leading to incorrect responses. In conclusion, the reliability of LLMs' internal knowledge is questionable, highlighting the need for clean and precise contextual information for generating factual responses.

GPT Family Steadily Improved Factuality We present the detailed results and rankings of the seven evaluated LLMs in Figure 12. We rank the LLMs based on the contradiction rates, where a lower contradiction rate is considered better. Across all three settings within our benchmark, a consistent enhancement in factuality is evident, progressing from InstructGPT to ChatGPT and to GPT-4.

Open Source LLM is Catching Up The analysis presented in Figure 12 also reveals noteworthy trends regarding the performance of open-source LLMs. Notably, proprietary LLMs, particularly GPT-4 and Claude 2, exhibit superior performance compared to open-source LLMs, on average. It is worth highlighting, however, that the recently open-sourced Llama 2 model demonstrates exceptional performance. Notably, in all evaluated settings, Llama 2 70B Chat outperforms ChatGPT and even surpasses Claude 2 in the Noisy Context setting. These findings suggest a potential for open-source LLMs to narrow the performance gap with their proprietary counterparts.